



MATHEMATICS METHODS : UNITS 1 & 2, 2019

Test 2 – (10%)

(1.1.1 – 1.1.28)

Time Allowed	First Name	Surname	Marks
20 minutes			25 marks

Circle your Teacher's Name:

Mrs Benko

Mrs Bestall

Mrs Fraser-Jones

Mrs Goh

Mr Koulianos

Mrs Murray

Mr Rudland

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

CALCULATOR FREE

1. [3 and 3 marks]

Solve:

a. $\frac{5y+1}{4} = 6 - \frac{2y}{3}$

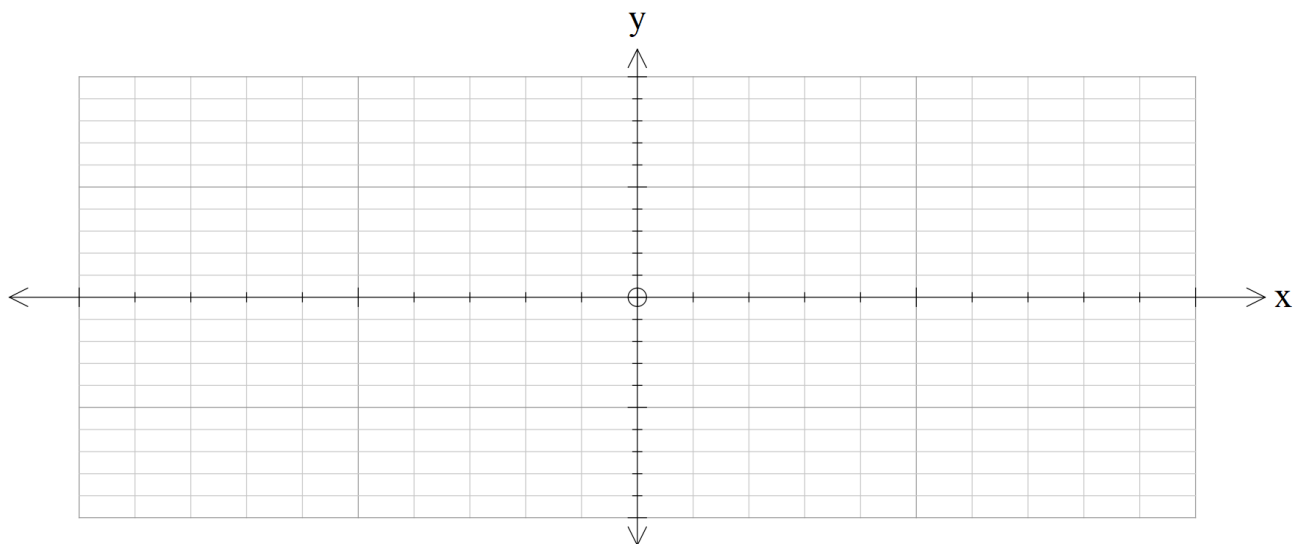
b. $x^2 + 2x = 15$

2. [4 and 5 marks]

Given $x^3 + 3x^2 - 18x - 40 = (x + a)(bx^2 + cx - 8)$

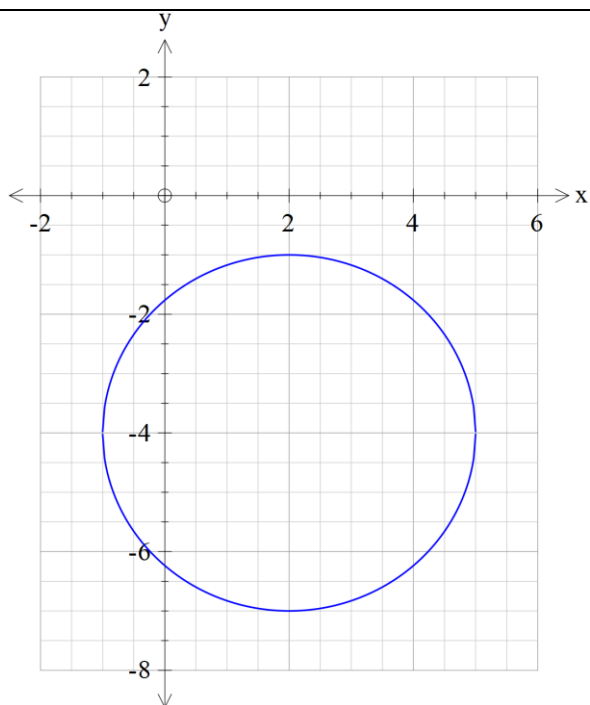
a. Find the value of a , b and c .

b. Hence sketch $y = 40 + 18x - 3x^2 - x^3$, showing x and y intercepts.



3. [10 marks]

Using the graph supplied, find the value of the constants in each of the equations below:

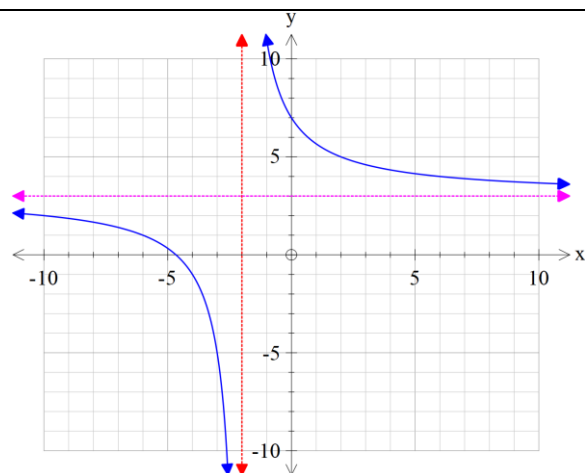


$$(x + a)^2 + (y + b)^2 = c$$

a =

b =

c =

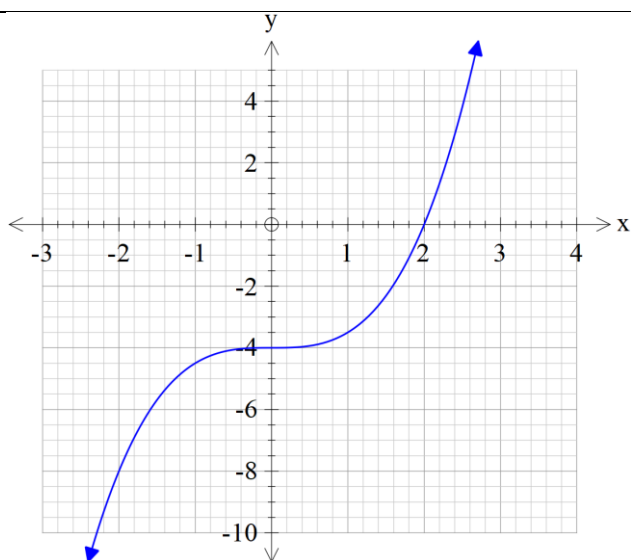


$$y = \frac{d}{x + e} + f$$

d =

e =

f =

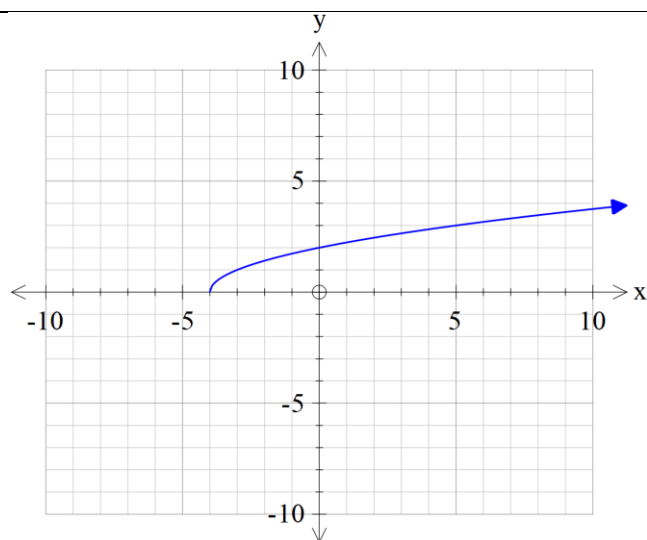


$$y = g(x + h)^3 + i$$

g =

h =

i =



$$y = \sqrt{x + j}$$

j =

Reading Time: 1 minutes



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Time Allowed	First Name	Surname	Marks
25 minutes			29 marks

Circle your Teacher's Name: Mrs Benko Mrs Bestall Mrs Fraser-Jones
 Mrs Goh Mr Koulianos Mrs Murray Mr Rudland

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

CALCULATOR Assumed

1. [2 marks]

A relation consists of the ordered pairs: $(-3,4)$, $(-1,5)$, $(0,-2)$, $(1,4)$ and $(6,8)$. Is the relation a function? Explain.

2. [5 marks]

Express $y = 4x^2 + 8x - 7$ in the form $y = a(x + b)^2 + c$ and hence give its domain and range.

3. [3 marks]

The stress on an object is inversely proportional to its area. If a rectangle measuring 4 metres by 6 metres is under a stress of 60 N/m^2 . Find the stress a 3 m square would encounter with the same force.

4. [4 marks]

The equation of the path of a cricket ball is $y = 1.1x - \frac{x^2}{50}$ where x and y are the horizontal distance travelled and the vertical height respectively in metres. Find, to the nearest the greatest vertical height reached and the horizontal distance travelled.

5. [3 marks]

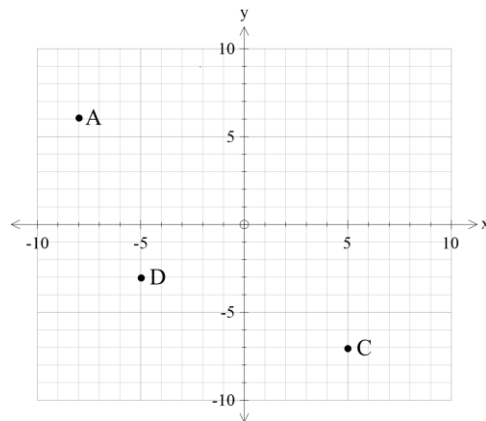
Find the x-intercept of the line parallel to $5x - 2y + 10 = 0$ with a y intercept of (0,-7).

6. [3 marks]

Find the value(s) of m if the quadratic equation $x^2 - 2mx + 3 = 0$ has only one solution.

7. [1, 3, 3 and 2 marks]

$A(-8,6)$, $B(a,b)$, $C(5,-7)$ and $D(-5,-3)$ are the vertices of a quadrilateral with diagonals AC and BD .



- Find the mid-point of AC .
- Find the co ordinates of B , given the diagonals of $ABCD$ bisect each other.
- Find the equation of the line perpendicular to DC , and passing through D .
- Does B lie on the line found in question c.